1. Meta Information

Project title: District-Level Multidimensional Poverty and Health in India

Course: CS215 – Data Analysis and Interpretation

Group ID: 08

Mentor: Namish Bansal

Team members:

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- Arshdeep Singh – 25B1082
- LLM

Honor code and external discussions: The team reports that it did not discuss the project with people outside the group.

LLM usage:

- https://chatgpt.com/share/6aeb5a5a-0b28-83ee-aa47-63a72b5b7ee9
- https://chatgpt.com/share/6aeb647c-bc70-83ee-8102-9ec708ee3c63
- https://chatgpt.com/share/6aeb65ae-e014-83ee-95fb-c1b0c2b8bb55
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- https://chatgpt.com/share/6acd30e7-df5c-83ec-a97d-7e45aa08ebbe
- https://chatgpt.com/share/6aeb82cc-c5f8-83ee-894f-17b9934bd05c
- https://chatgpt.com/share/6aeb82d9-ddf4-83ee-ab2b-f3a808bb889c
- https://chatgpt.com/share/6aeb8323-cf0c-83e8-99e1-ffdb5850f54d
- https://claude.ai/share/cbdf46e7-f670-4ad1-803f-12c363645bb1
- https://share.gemini.google/1P00cR2BdnHd

Video presentation: https://drive.google.com/file/d/18dAygKOFWkxBjdl62zrgdE1exS2HCG8f/view?usp=share_link/

GitHub repository: github.com/rishabhmeel/dai-project-district_analysis

GitHub handles and earliest project commits:

- Rishabh Meel: rishabhmeel
- Sugam Kumar Thakur: Sugam_kumar
- Aaryan Chavda: AaryanChavda01
- Arshdeep Singh: arshhhhhhhhh
  Earliest meaningful commit: Oct 19, 2026

Contribution statement: Contributions were roughly equal. Team members worked both independently and collaboratively throughout the project. We would be giving 100% scores to everyone.

2. Technical Report

2.1 Problem and motivation

This project investigates the relationship between district-level multidimensional poverty and selected health indicators in India. Its central research question is: *How strongly, and in what direction, is the Multidimensional Poverty Index (MPI) associated with health outcomes across Indian districts, and which indicators show the strongest associations?*

MPI summarizes deprivation across multiple dimensions. Comparing it with health indicators can help describe how poverty and health outcomes vary together across districts. The analysis is exploratory and observational: it identifies statistical associations, not causal effects.

2.2 Data sources and structure

The analysis combines district-level MPI data with health indicators from the National Family Health Survey (NFHS-5), collected during 2019–2021. The merged dataset contains 707 districts and eight health indicators.

The data-loading workflow reads the source spreadsheets, selects the required variables, harmonizes district names, and merges the datasets. Name harmonization accounts for spelling differences and district renaming. The cleaned merged dataset is then used by the analysis notebook.

The health indicators are percentages. They cover menstrual hygiene, improved sanitation, full vaccination, adequate diet among children aged 6–23 months, anaemia among women aged 15–49, high blood sugar among men, tobacco use among women, and alcohol use among men.

2.3 Data preparation

District names and identifiers were standardized before merging. Values represented as negative numbers for parenthesized estimates were converted to positive percentages, while suppressed values were treated as missing. Missing values were not imputed.

The final dataset includes 707 districts. Full vaccination has 685 available observations and adequate diet has 706; the other listed variables have 707 observations. Pairwise complete observations were used for the relevant correlations and regressions.

2.4 Tools and analysis workflow

The project was conducted in Python using Jupyter Notebook. The main libraries were pandas, NumPy, Matplotlib, and SciPy.

The workflow consisted of:

- loading and merging the source datasets;
- checking missing values and descriptive statistics;
- plotting the distribution of district MPI;
- calculating Pearson and Spearman correlations for each health indicator;
- visualizing the two strongest negative associations using scatterplots and fitted trendlines;
- fitting simple linear regressions for menstrual hygiene and improved sanitation.

Pearson's r measures linear association, while Spearman's ρ measures rank-based monotonic association. Regression slopes describe the fitted change in an indicator associated with a change in MPI; they should not be interpreted as causal effects.

2.5 Descriptive statistics

Table 1 summarizes MPI and the eight health indicators. The mean district MPI was 0.065, with a median of 0.053. The distribution is right-skewed: most districts have relatively low MPI values, while a smaller number

have substantially higher values.

Table 1: Descriptive statistics for MPI and health indicators

Variable	Mean	Std. dev.	Median	N
MPI	0.065	0.056	0.053	707
Improved sanitation	71.917	14.283	73.730	707
Menstrual hygiene	78.271	15.228	79.600	707
Full vaccination	84.033	10.020	84.970	685
Adequate diet	12.329	7.485	10.920	706
Anaemia (women)	55.944	11.966	57.180	707
High blood sugar (men)	14.728	4.952	14.110	707
Tobacco use (women)	11.615	11.943	7.670	707
Alcohol use (men)	23.191	13.362	20.160	707

The health indicators vary substantially across districts. For example, improved sanitation ranges from 29.24% to 99.85%, while menstrual hygiene ranges from 8.72% to 100%. Tobacco use among women also has a wide range, from 0.06% to 70.58%.

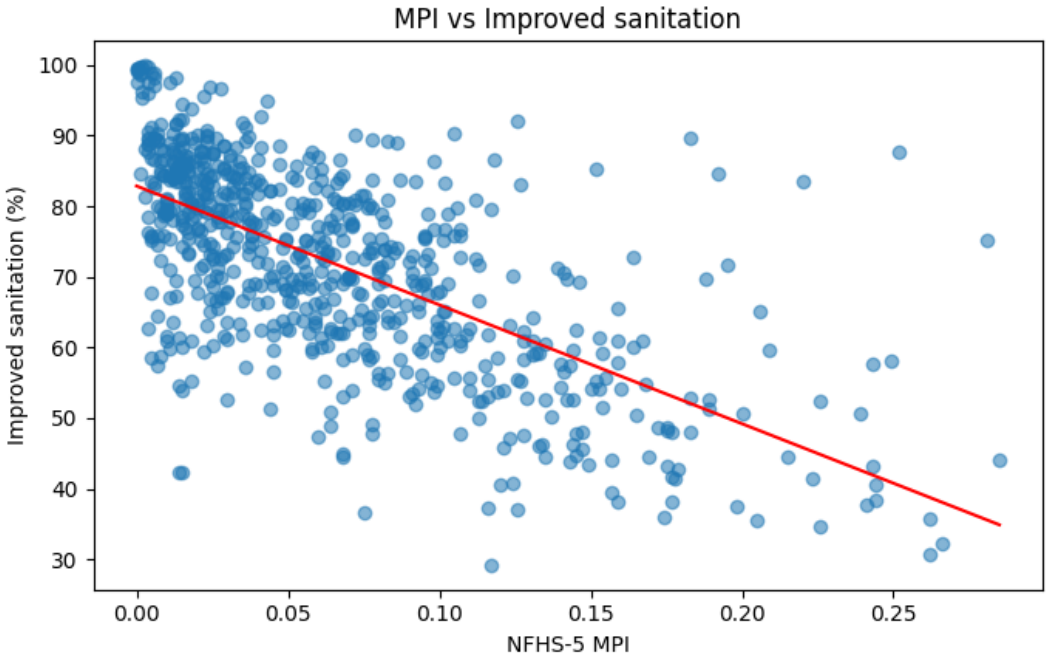


Figure 1: Distribution of district MPI. The distribution is concentrated at lower MPI values with a right tail toward higher deprivation.

2.6 Correlation results

Table 2 reports Pearson correlations, p-values, and Spearman rank correlations. The strongest negative associations with MPI are for menstrual hygiene and improved sanitation. Full vaccination, high blood sugar among men, and adequate diet also have negative correlations, but their magnitudes are smaller.

Table 2: Correlation of MPI with health indicators

Indicator	N	Pearson r	Spearman ρ
Menstrual hygiene	707	-0.748	-0.795
Improved sanitation	707	-0.656	-0.640
Full vaccination	685	-0.324	-0.370
High blood sugar (men)	707	-0.314	-0.374
Adequate diet	706	-0.179	-0.259
Alcohol use (men)	707	-0.025	-0.102
Tobacco use (women)	707	0.222	0.364
Anaemia (women)	707	0.242	0.268

The Pearson p-value for alcohol use among men is 0.511, so its linear association with MPI is not statistically significant at conventional levels. The other listed Pearson correlations have p-values below 0.001.

The positive correlations for anaemia and tobacco use indicate that districts with higher MPI tend to have higher reported prevalence of these indicators. This is an association across districts and does not imply that MPI directly causes either outcome.

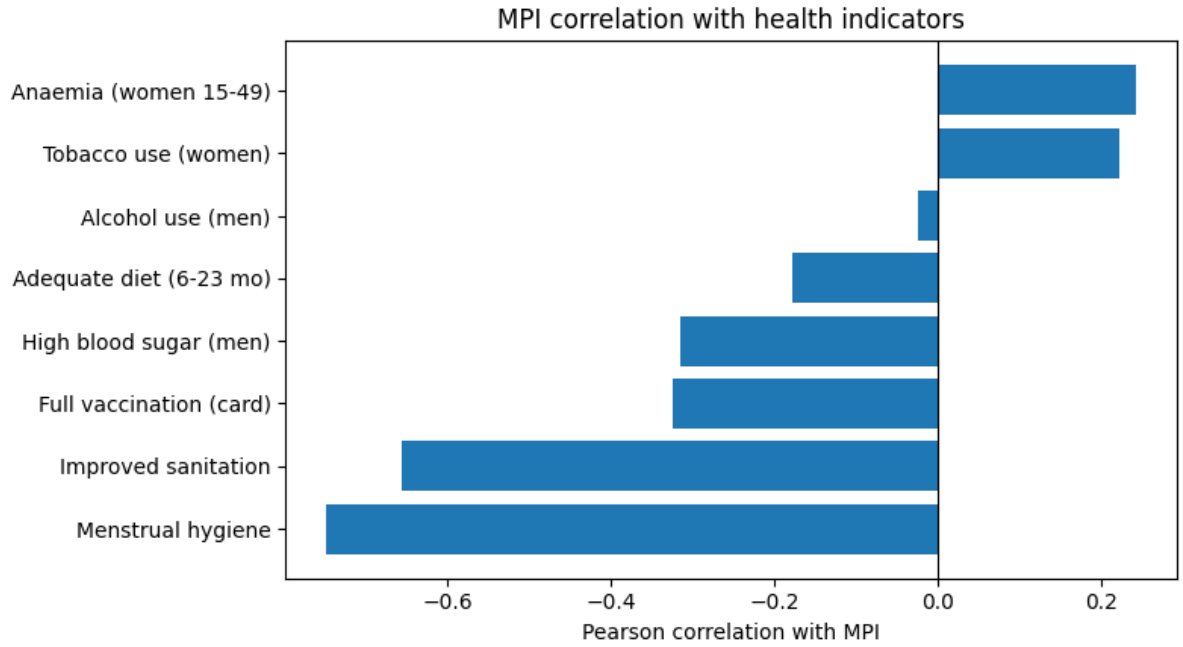


Figure 2: Pearson correlations between MPI and the eight health indicators. Bars to the left indicate negative associations; bars to the right indicate positive associations.

2.7 Regression and scatterplot analysis

Simple linear regressions were fitted for menstrual hygiene and improved sanitation, the two indicators with the strongest negative Pearson correlations.

For menstrual hygiene, the fitted slope is −204.637 percentage points per 1.0 MPI, equivalent to an estimated decrease of 20.464 percentage points for a 0.1 increase in MPI. The intercept is 91.566 and $R^2 = 0.559$.

For improved sanitation, the fitted slope is −168.345 percentage points per 1.0 MPI, equivalent to an estimated decrease of 16.835 percentage points for a 0.1 increase in MPI. The intercept is 82.855 and $R^2 = 0.430$.

These fitted relationships summarize the observed district-level patterns. The slopes should not be extrapolated beyond the observed MPI range or interpreted as the effect of changing poverty in a district.

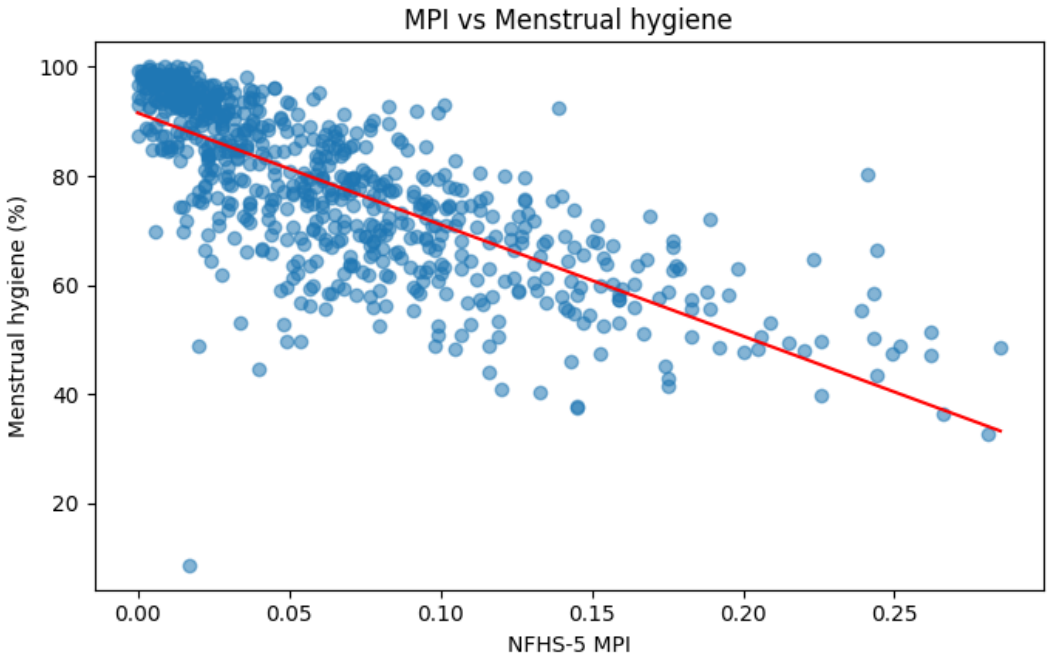


Figure 3: MPI versus menstrual hygiene. The fitted line shows a strong negative linear association across districts.

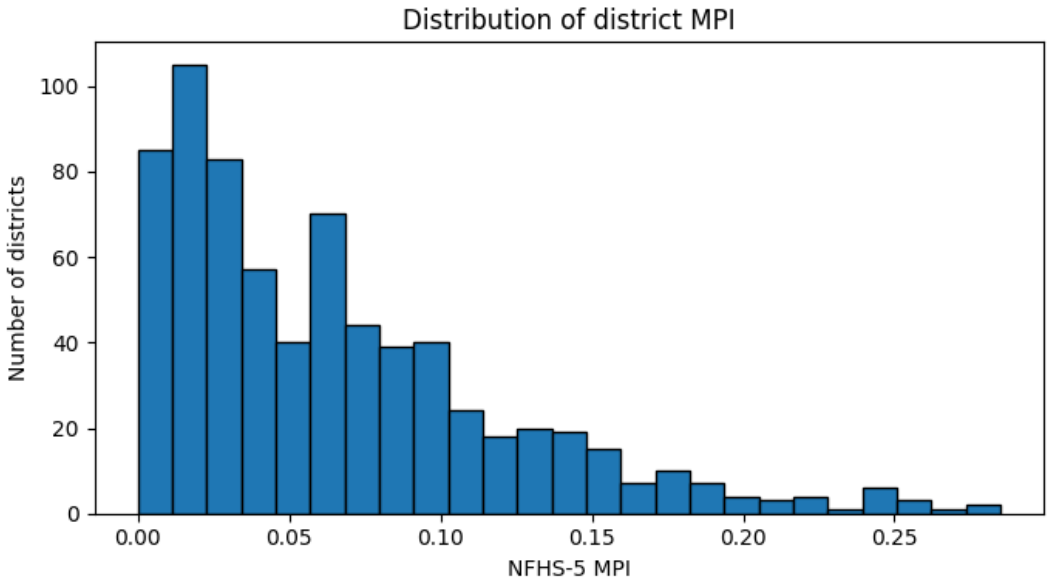


Figure 4: MPI versus improved sanitation. The fitted line shows a negative association, with substantial variation around the line.

2.8 Challenges and limitations

A key data-preparation challenge was harmonizing district names across sources, including spelling differences and renamed districts. Some health-indicator values were also suppressed or specially encoded, requiring consistent handling.

The analysis is observational and district-level, so it cannot establish causality or describe individual-level relationships. District averages may hide substantial within-district variation. Regional, demographic, and socioeconomic factors may influence both MPI and health outcomes. Sanitation is also represented within MPI, so its association with MPI may partly reflect conceptual overlap. Finally, the data describe a specific survey period and may not represent current conditions.

2.9 Conclusion

The analysis finds that district MPI is most strongly negatively associated with menstrual hygiene and improved sanitation. Associations with vaccination, high blood sugar, and adequate diet are weaker, while anaemia and tobacco use among women show positive associations. Alcohol use among men has little linear association with MPI in this dataset.

Overall, MPI is associated with several important health indicators, but the strength and direction differ by indicator. These results provide a descriptive comparison across districts and do not establish causal relationships.

3. Frequently Asked Questions

Q1. Which health indicators are most strongly associated with MPI, and how large are these relationships?

Menstrual hygiene has the strongest negative Pearson correlation with MPI ($r = -0.748$, $N = 707$), followed by improved sanitation ($r = -0.656$, $N = 707$). Their Spearman correlations are also strong and negative ($\rho = -0.795$ and -0.640), supporting the conclusion that districts with higher MPI generally rank lower on these indicators. These are the clearest associations in the analysis, though they do not establish causation.

Q2. Are the associations with vaccination, high blood sugar, and adequate diet comparable to those for sanitation and menstrual hygiene?

No. Full vaccination has a moderate negative correlation with MPI ($r = -0.324$, $N = 685$), as does high blood sugar among men ($r = -0.314$, $N = 707$). Adequate diet has a weaker negative correlation ($r = -0.179$, $N = 706$). Their magnitudes are smaller than those for sanitation and menstrual hygiene, suggesting that the association with MPI differs substantially across health indicators.

Q3. Which indicators increase with MPI, and how should these positive correlations be interpreted?

Anaemia among women ($r = 0.242$, $N = 707$) and tobacco use among women ($r = 0.222$, $N = 707$) have positive correlations with MPI. This means that districts with higher MPI tend to have higher reported prevalence of these indicators. The associations are relatively modest compared with the negative correlations for menstrual hygiene and sanitation, and they do not show that MPI causes either outcome.

Q4. Is alcohol use among men meaningfully associated with MPI in this dataset?

The Pearson correlation is $r = -0.025$ ($N = 707$), with a p-value of 0.511. The coefficient is very close to zero, and the p-value does not provide evidence of a linear association at conventional significance levels. Spearman's correlation is also small ($\rho = -0.102$), so the data show little rank-based association as well. This does not prove that no relationship exists; it indicates that the observed association is weak in this sample.

Q5. What does the difference between Pearson and Spearman correlations reveal about tobacco use among women?

For tobacco use among women, Pearson's correlation is 0.222, while Spearman's correlation is larger at 0.364. This suggests that the districts' rank ordering in MPI and tobacco use is more consistently increasing than a straight-line relationship between their original values would indicate. The difference is a reason to avoid describing the relationship as purely linear; the coefficients alone do not identify its exact functional form.

Q6. What do the regression results show for menstrual hygiene and improved sanitation?

For menstrual hygiene, the fitted slope is -204.637 percentage points per 1.0 MPI, equivalent to an estimated decrease of 20.464 percentage points for a 0.1 increase in MPI. The model has $R^2 = 0.559$, meaning the fitted line accounts for about 55.9% of the observed district-level variation in menstrual hygiene. For improved sanitation, the corresponding estimated decrease is 16.835 percentage points per 0.1 MPI, with $R^2 = 0.430$. Both models show substantial negative associations, but the remaining variation indicates that MPI alone does not account for all district differences.

Q7. How should the descriptive statistics help interpret the correlation results?

MPI has a mean of 0.065, a median of 0.053, and a standard deviation of 0.056, with values ranging from 0 to 0.285. The histogram shows that many districts are concentrated at lower MPI values, with a right tail toward more deprived districts. Menstrual hygiene and sanitation vary widely across districts, with standard deviations of 15.228 and 14.283 percentage points, respectively. This variation provides context for the observed associations, while the skewed MPI distribution cautions against assuming a uniform relationship across the full range.

Q8. What conclusions are supported by the analysis, and what remains uncertain?

The data support the conclusion that MPI is strongly negatively associated with menstrual hygiene and improved sanitation, moderately negatively associated with vaccination and high blood sugar among men, and weakly negatively associated with adequate diet. Anaemia and tobacco use among women have modest positive associations, while alcohol use among men has little observed association. Because the data are observational and aggregated by district, the analysis cannot determine causal effects or establish whether the same patterns hold for individuals or in other time periods.

4. Reflection

This open project gave us an opportunity to apply data analysis to a real-world question rather than work with a dataset and procedure prescribed in advance. We had to define a research question, identify relevant data, understand how the datasets were structured, and decide which statistical methods could help answer the question. This required more than implementing formulas: we had to make and justify decisions throughout the analysis.

One of our main takeaways was the importance of data preparation. Combining district-level MPI and health data required us to reconcile district names, account for renamed districts, and handle missing or specially encoded values consistently. These tasks were less visible in textbook examples, but they were essential to making the analysis reliable. We also gained practical experience using Python and Jupyter Notebook to organize a workflow, inspect data, calculate descriptive statistics, visualize distributions, and compare Pearson and Spearman correlations.

The results encouraged us to think beyond whether a correlation was positive or negative. For example, MPI had stronger negative associations with menstrual hygiene and improved sanitation than with vaccination or adequate diet. The differences between Pearson and Spearman correlations also showed that a single statistic may not fully describe a relationship. Interpreting these results required us to consider the meaning of each indicator, the structure of the data, and the limits of district-level comparisons. We learned to distinguish an observed association from a causal explanation.

The open-ended nature of the project was challenging at first because there was no fixed sequence of steps or guaranteed correct answer. As the work progressed, that uncertainty became an opportunity to take ownership of the problem, divide tasks, compare approaches, and refine our analysis through discussion. Working both independently and collaboratively helped us make progress while keeping the overall project coherent.

Compared with a standard homework assignment, the project required a broader and more sustained effort. The work involved not only analysis, but also data sourcing, cleaning, documentation, interpretation, and communication of findings. These are relevant skills for further academic work and for industry settings, where data is often imperfect and the problem itself may not be clearly defined at the outset.

Overall, the project strengthened our understanding of the complete data-analysis process: from turning a broad question into an analysis plan to presenting conclusions with appropriate caution. It also reinforced that the value of an analysis depends not only on the results it produces, but on the quality of the data, the reasoning behind the methods, and the care taken in interpreting what the results can show and cannot show.